

Adaptive Traffic Signal Control Based on A Modified Proximal Policy Optimization Algorithm

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Abstract—Reinforcement learning has been shown to effectively optimize traffic signal timing. This study focuses on reducing vehicle fuel consumption and introduces an adaptive traffic signal control model, termed MPPO_TSC. This model incorporates various algorithmic enhancements over the standard Proximal Policy Optimization (PPO) framework and integrates both Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to improve the capability of deep neural networks in extracting features from dynamic traffic flows. Additionally, to overcome the challenge of agents being unable to access real-time vehicle fuel consumption data, this study devises an approximate method for estimating differential fuel consumption and develops a novel reward function accordingly. Experimental validation was carried out on the SUMO simulation platform, with tests conducted in both simulated and real-world intersection scenarios. The results indicate that MPPO_TSC achieves reductions in average fuel consumption per vehicle and decreases in average waiting time compared to the four control strategies.

Keywords—traffic signal control; deep reinforcement learning, modified Proximal Policy Optimization, fuel consumption.

I. INTRODUCTION

In the field of intersection signal control, traditional methods mainly include fixed-time control and automatic induction control. With the continuous development of the internet of vehicles and artificial intelligence derived from the internet, more and more researchers are using machine learning methods for adaptive control of traffic signals. Reinforcement Learning can be well modeled as a Markov Decision Process (MDP) [1]. Mao et al. [2] deployed the Soft Actor-Critic (SAC) algorithm for traffic signal management, evaluating it alongside other reinforcement learning strategies. To broaden applicability, Boukerche et al. [3] utilized the Deep Q-learning (DQN) algorithm, optimizing average waiting times, queue lengths, and delays as metrics to concurrently reduce fuel consumption and vehicular transit times. Zhang et al. [4] crafted an adaptive traffic signal priority model that prioritizes efficiency improvements for buses and general vehicles, thereby decreasing their waiting times. Jiang et al. [5]

introduced a reinforcement learning framework for traffic signal control under general unknown scenarios, enhancing the model's adaptability. Tan et al. [6] proposed a novel data-driven approach for timing control, while Vlachogiannis et al. [7] developed a decentralized adaptive traffic signal control algorithm, optimizing pedestrian flow at intersections. Gu et al. [8] introduced a regional signal control framework that enables agents to efficiently search for the optimal actions. Tan et al. [9] offered an adaptive traffic signal control approach with integrated privacy protection, considering the privacy needs of connected vehicle passengers. Existing researches have made significant progress in the field of adaptive traffic signal control, but there are problems such as low utilization of algorithms, insufficient state feature extraction capability and action flexibility, and inconvenient access to vehicle fuel consumption data. More importantly, existing researches still lack an in-depth study of the adaptive signal control method of deep reinforcement learning from the perspective of fuel consumption. This paper will address the aforementioned shortcomings.

II. METHODOLOGY

The Proximal Policy Optimization[10] algorithm employs importance sampling to ensure that samples from the old policy remain relevant for updates under the new policy. The importance sampling ratio, indicating the relative probability between the new and old policies, is calculated as:

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)} = \frac{p_{\theta}(s_t, a_t)}{p_{\theta}(s_t, a_t)} \quad (1)$$

Here, $r_t(\theta)$ signifies the importance sampling ratio, where $\pi_{\theta}(a_t | s_t)$ is the probability of selecting action a_t under the new policy.

To ensure stable policy updates, PPO's core formula incorporates a clipping mechanism. The specific policy gradient loss function is articulated as:

$$L^{PG}(\theta) = E_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)] \quad (2)$$

The overall objective function in PPO amalgamates the policy gradients, value function loss, and the policy's entropy into a cohesive multi-component objective, formalized as:

$$L_t(\theta) = E_t[L_t^{PG}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t)] \quad (3)$$

III. MODELING

A. States

For states, we employ discrete traffic state encoding (DTSE), which excels in discretizing the traffic state environment. This method is not only compatible with the proposed neural network model but also enriches the state information available to the agent. By transforming the location and speed data of vehicles into an image format, the DTSE approach facilitates the neural network's processing of this data. Furthermore, the area around the intersection is segmented into multiple small grids, forming matrices for position and speed data. Each cell in the position matrix is marked with a 1 if a vehicle is present and 0 otherwise. Similarly, cells in the speed matrix contain speed values in meters per second (m/s) for each vehicle present, or 0 if no vehicle is present.

B. Action Space

The model in this study is designed for a typical four-phase intersection with separate lanes for left turns, through traffic, and combined straight-right turns. Vehicles in the left turn and through lanes are permitted to move only during their respective green lights, while right-turning vehicles can proceed when unimpeded by through traffic. Taking into account the patience of drivers waiting at red lights and pedestrian crossing needs, each phase is assigned maximum and minimum green light durations, with yellow lights included during transitions for added safety. The control model defines the phase set as {EWG, EWLG, NSG, NSLG} and the action set as $A=\{0,1\}$. Here, action 0 maintains the current phase, while action 1 switches to the next phase.

C. Rewards

This paper introduces a novel approach for approximating differential fuel consumption. The method for approximating idle fuel consumption involves multiplying the total amount of time a vehicle spends idling by its rate of fuel consumption during idling periods. As the engine power of a vehicle remains relatively constant while waiting at a red light, this idle fuel consumption rate f_{idle} , is typically treated as a fixed constant. The total idle time can be estimated used an approximation approach specifically designed for such periods. The computation for this accumulated time is as follows:

$$D_t = \frac{1}{2} * (q_t + q_{t+1}) * \Delta_t \quad (4)$$

Where t denotes time, n denotes the number of vehicles, q_t denotes the length of the vehicle queue at any given moment t , and Δ_t the duration of the executed action a .

Consequently, the method for calculating idle fuel consumption is specified as follows:

$$F_{idling} = \sum_{l=1}^n D_l * f_{idle} \quad (5)$$

The total displacement S_{total} experienced by a vehicle during its deceleration and acceleration phases can be calculated by :

$$S_{total} = (v_0 * t_1 + \frac{1}{2} * a_1 * t_1^2) + (v_1 * t_2 + \frac{1}{2} * a_2 * t_2^2) \quad (6)$$

The additional fuel consumption attributable to a single stop is then calculated by:

$$F_{stop} = (f_{acce} * t_1 + f_{dece} * t_2) - \frac{S_{total}}{V_{normal}} * f_{normal} \quad (7)$$

F_{stop} is multiplied by the number of stops, as outlined in :

$$F_{driving} = F_{stop} * veh_{halted_num} \quad (8)$$

This study introduces a novel reward function derived from the method of approximating differential fuel consumption. The function is as follows:

$$R = - \sum_{i=1}^n (F_{idling,i} + F_{driving,i}) \quad (9)$$

IV. EXPERIMENTS AND EVALUATION

A. Simulation Settings

This study examines a typical four-phase intersection featuring four entry and exit lanes. For simulation purposes, both the entry and exit lanes are 500m long, with a state collection area defined by a 150m perimeter around the traffic lights, covering a square region of 300m by 300m. The grid, used for simulation, measures 5m per unit, creating a total grid area of 60×60 units. The traffic simulation software SUMO is utilized to establish a single-point intersection as the experimental environment. Experiments are conducted using the PyTorch deep learning framework on NVIDIA RTX 3070Ti.

Regarding vehicle parameters, each vehicle in experiments measures 5m in length with a minimum safety distance of 2m maintained between vehicles. The Krauss following model dictates a maximum speed of 13.89m/s, with acceleration capped at 1m/s² and deceleration at 4.5m/s². The fuel consumption coefficient for vehicles in idle is set at 837mg/s, the additional fuel consumed during a stop is calculated as 15196mg. The RL hyperparameters are defined as shown in Table I.

B. Model Effectiveness

1) Estimation of Reward Approximation

Here, our objective is to assess the effectiveness of the differential fuel consumption approximation method introduced in this paper. The evaluation involves comparing the convergence behaviors of two different reward functions. The first Reward_1 is derived by aggregating the vehicle fuel consumption data collected through the TraCI interface integrated with SUMO, i.e. the traditional total fuel consumption reward function. The second, Reward_2, is based on the differential fuel consumption approximation method developed for this study. All experimental conditions remain consistent across the two setups, except for the reward function definitions. Fig. 1 illustrates the comparative analysis of the convergence behaviors of the PPO algorithm under these two

reward functions. Both Reward_1 and Reward_2 demonstrate converging trends post-training, displaying similar patterns of increase and decrease. Notably, the training with Reward_2 exhibits a smaller range of fluctuation, indicating a more stable control performance. While the curves converge, they do not overlap but maintain a consistent gap. This gap underscores the constant difference typically observed between differential and total fuel consumption in actual driving scenarios, validating that utilizing differential fuel consumption as the reward function not only mirrors the objectives achieved using total fuel consumption but also enhances the convergence performance of the model. Combined with the analysis of Fig. 1, we know that the specific advantage of Reward 2 over Reward 1 is that Reward 2 can reflect the increase or decrease in vehicle fuel consumption due to the efficiency of signal control, while Reward 1 can only see the overall fuel consumption of all vehicles. Therefore, Reward 2 is more helpful for the Agent to flexibly regulate the signal timing scheme. And performance comparison of Reward 2 proposed in this paper in different traffic scenarios(simulation and Jining scenarios) is shown in Table II and Table III.

TABLE I. Setting of Hyperparameters

Hyperparameter	Value
Lr (α)	3e-4
Gamma (γ)	0.99
Clipping ratio	0.2
Batch size	128
Minibatch size	16
Lambda (λ)	0.95
K_epochs	20
c_1	0.5
c_2	0.01

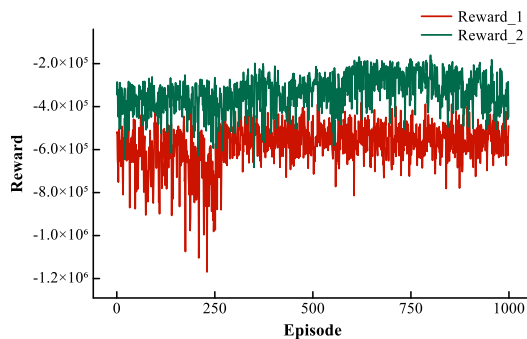


Fig. 1. Comparison of reward functions.

2) Algorithm Tricks

This experiment will validate the proposed algorithmic tricks. Firstly, the implementation of all the tricks is analyzed. In this paper, tricks such as state normalization, advantage normalization, adaptive decay of learning rate, orthogonal initialization of neural network weights, and separate setting of Adam optimizer parameters are introduced to the standard PPO algorithm. In the training process of the model, the weights of the network model are first initialized using the orthogonal initialization method, and then the states are normalized to ensure the ability of the neural network to extract the environmental feature information. Subsequently, the technique

of dominance normalization is used in the process of calculating the generalized dominance function of the PPO, and the Epsilon parameter of the Adam optimizer is individually set to 1e-5, and finally, the technique of adaptive decay of the learning rate is executed during the training process in order to ensure that the intelligent body learns and utilizes the optimal strategy. Secondly to illustrate its specific impact on the performance of the algorithm, Fig. 2 illustrates the comparison between whether the trick is introduced or not, from which it can be seen that after the introduction of the trick, the PPO obtains a better convergence and gradually stabilizes in the later stages of the training. In contrast, without tricks, the PPO always fluctuates and converges poorly. This shows that the improvement of the PPO algorithm in this paper is effective and necessary.

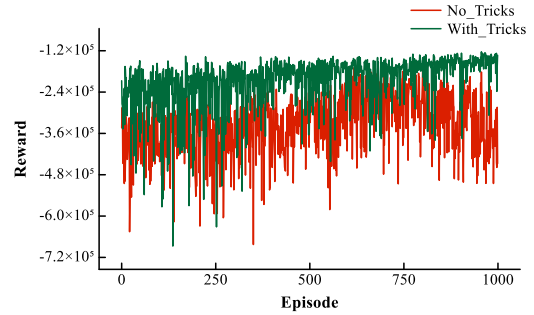


Fig. 2. Comparison of algorithm tricks.

3) Improvement of Network Architecture

Fig. 3 illustrates the comparative training trajectories of the PPO algorithm using different neural network architectures. The CNN exhibits slow convergence amidst considerable fluctuations and achieves the least effective convergence. In contrast, Fully connected (FC) shows a gradual convergence throughout its training, albeit with persistent significant fluctuations. On the other hand, CNN-LSTM starts to converge effectively after approximately 400 training rounds, demonstrating a superior convergence quality. Concerning algorithmic stability, CNN, despite being relatively stable towards the end of the training, performs poorly in control, failing to meet practical requirements. FC, while outperforming CNN in control performance, consistently experiences fluctuations, suggesting that the agent has not yet mastered an optimal strategy for the current traffic scenario, thus failing to stabilize control. CNN-LSTM, commencing from round 400, adopts an effective control strategy with minimal fluctuation, indicating sustained stable control. The superior performance of CNN-LSTM confirms the efficacy of our enhancements to the PPO's neural network architecture.

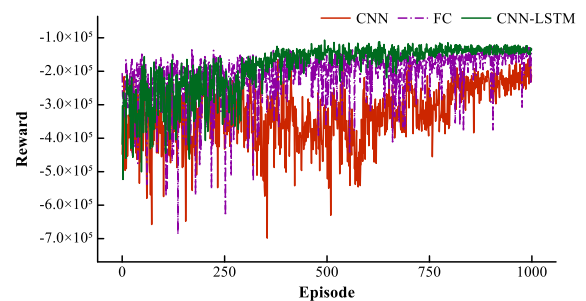


Fig. 3. Comparison of neural network architectures.

C. Model Comparison

The comparative results of these different signal control methods are presented in Fig. 4. Upon analyzing the test results with Table II, the average fuel consumption (AFC) metrics are as follows: Fix time control (FTC) recorded 120ml, Actuated Traffic Signal Control (ATSC) 116ml, D3QN-Waiting 110ml, D3QN-Fuel 102ml, SAC-Waiting 101ml, SAC-Fuel 99ml, and MPPO 93ml. Initially, we examined the effects of different reward functions on D3QN and SAC. D3QN, when optimizing for fuel consumption, achieved a 7.3% reduction in AFC compared to its performance under the D3QN-Waiting setup. Similarly, SAC-Fuel showed a 2% improvement over SAC-Waiting. We then assess MPPO_TSC's fuel consumption performance relative to other signal control models. Compared to FTC, MPPO_TSC achieves a 22.5% reduction in AFC; it shows a 19.8% reduction compared to ATSC, 15.5% compared to D3QN-Waiting, 8.8% compared to D3QN-Fuel, 7.9% compared to SAC-Waiting, and 6.1% compared to SAC-Fuel. Although MPPO demonstrates an advantage in the AFC metric, relying on a single metric is insufficient to capture the algorithm's overall performance. Therefore, a multi-metric evaluation is necessary to fully assess the algorithm's capabilities. Fig 5 illustrates the average waiting time (AWT) metrics for all methods under the test scenarios, showing that reinforcement learning-based control strategies generally outperform FTC and ATSC, with MPPO and SAC achieving the best results. As indicated in Table III, the AWT for FTC is 39.61s, ATSC is 33.45s, D3QN-Fuel is 31.14s, SAC-Fuel is 29.46s, and MPPO is 29.08s. From these comparisons, it is evident that MPPO reduces AWT by 26.6% compared to FTC, 13.1% compared to ATSC, 6.6% compared to D3QN, and 1.3% compared to SAC. These results suggest that MPPO continues to offer strong performance in terms of AWT control.

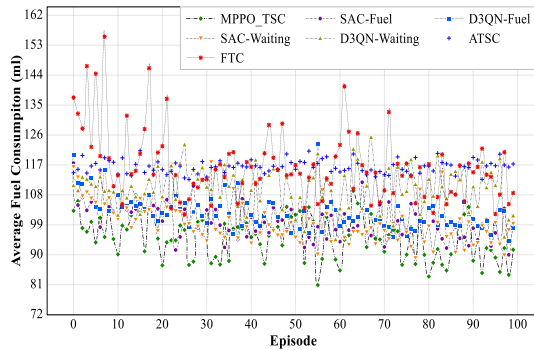


Fig. 4. Comparison of AFC for all control strategies.

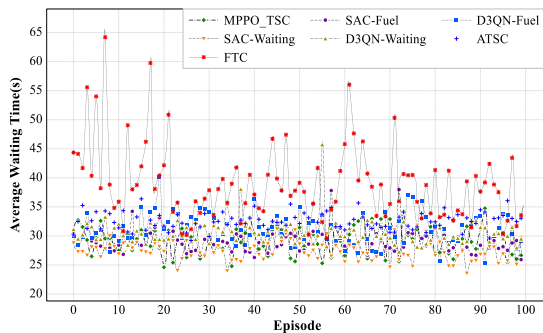


Fig. 5. Comparison of AWT for all control strategies

TABLE II. Comparison of AFC Across All Control Strategies

Control Strategies	Traffic Scenario		Improvement
	Simulation Scenario	Jining Scenario	
FTC	128	120	22.5%
ATSC	120	116	19.8%
D3QN-Fuel	109	102	8.8%
D3QN-Wating	116	110	15.5%
SAC-Fuel	102	99	6.1%
SAC-Waiting	110	101	7.9%
MPPO	96	93	-

TABLE III. Comparison of AWT Across All Control Strategies

Control Strategies	Traffic Scenario		Improvement
	Simulation Scenario	Jining Scenario	
FTC	43.82	39.61	26.6%
ATSC	36.64	33.45	13.1%
D3QN-Fuel	33.31	31.14	6.6%
D3QN-Wating	32.49	30.28	4.0%
SAC-Fuel	30.24	29.46	1.3%
SAC-Waiting	29.33	28.77	-1.1%
MPPO	29.64	29.08	-

V. CONCLUSION

In this study, we enhance the standard PPO algorithm for TSC by integrating specific algorithmic techniques and refining the architecture of the deep neural network. The model processes state information such as vehicle position and speed, and in a fixed phase sequence, chooses actions to either maintain or change signal phases. Utilizing the proposed method for approximating differential fuel consumption, this metric serves as the reward function for the agent. The enhancements in algorithmic techniques significantly boost the control performance of the PPO algorithm, while improvements in the neural network architecture increase its capacity to discern features in complex traffic flows.

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